



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

KALOKI CARRICK MWELII
24/06590

SUPERVISOR – LISPER KENDI

SMART BANKLYTICS

An Intelligent Data – Driven Bank Management and Customer Analytics System



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

What We'll Cover

- Introduction & Background
- Problem Statement
- Objectives
- System Architecture & Tech Stack
- System Modules
- Fraud Detection
- Analytics Dashboard
- Testing Summary
- Challenges & Solutions
- Conclusion



INTRODUCTION & BACKGROUND

Background

Digital banking generates massive volumes of transaction data daily

Most small-scale banking systems focus only on basic transaction processing

Limited capabilities for analytics, fraud detection, and intelligent reporting

A clear gap exists between traditional banking and modern data-driven platforms

Smart Banklytics bridges this gap



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

PROBLEM STATEMENT

Many small and medium banking systems lack:

Integrated real-time analytics

Automated fraud detection

Intelligent financial reporting

Tools for monitoring customer behaviour

Result: Increased fraud risk, poor financial planning, and inefficient service delivery



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

OBJECTIVES

Main Objective To develop an intelligent banking system integrating transaction management, data analytics, and fraud detection using data science techniques.

Specific Objectives

Review and analyze transaction data to generate insights and detect suspicious activities

Design and implement a secure banking system for account management and transactions

Collect and store transaction data for analytical processing

Develop an interactive analytics dashboard for transaction trend visualization

Develop a rule-based fraud detection system for identifying suspicious transactions

Generate automated transaction reports and downloadable bank statements



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

SYSTEM ARCHITECTURE

Three-Tier Architecture

Layer	Component	Technology
Presentation Layer	Frontend UI	HTML, CSS, JavaScript
Application Layer	Backend Logic	Python, Flask
Data Layer	Database	SQLite

Flow: User → Web Interface → Flask Backend → SQLite Database → Response



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

TECHNOLOGY STACK

Tools & Technologies Used

Component	Technology
Backend	Python 3.x + Flask
Frontend	HTML, CSS, JavaScript
Database	SQLite
PDF Generation	fpdf Library
Data Analysis	Pandas, NumPy
Visualization	Chart.js
Platform	Web-based



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

SYSTEM MODULES

7 Core Modules

Module	Function
User Authentication	Secure login, sessions, role-based access
Customer Dashboard	Balance, transaction history, fraud alerts
Transaction Processing	Deposit, withdrawal, transfer
Fraud Detection	Automated rule-based monitoring
Admin Management	User control, fraud review, deposits
Analytics Dashboard	KPIs, charts, trends
PDF Statement Generator	Downloadable filtered bank statements



TRANSACTION PROCESSING

Three Transaction Types

Deposit Admin deposits funds directly into customer accounts → balance updated → transaction logged

Withdrawal Customer requests withdrawal → balance checked → fraud checked → balance deducted → logged

Transfer Customer initiates → receiver verified → balance checked → fraud checked → both accounts updated atomically → both transactions logged

All transactions pass through fraud detection before being committed



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

FRAUD DETECTION MODULE

Automated Rule-Based Detection

Rule 1 — Large Transaction Any single transaction exceeding **KES 100,000** → Account immediately suspended → Fraud alert created

Rule 2 — Rapid Transactions **3 or more transactions within 5 minutes** → Account immediately suspended → Fraud alert created

Admin Review Actions

- ✓ Legitimate → Transaction processed, account reactivated
- 🔍 Investigate → Alert flagged, account stays suspended
- ✗ Freeze → Account permanently frozen

Customer Appeal Customers can appeal alerts → admin notified for review



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

ANALYTICS DASHBOARD

KPI Metrics (Live from Database)

- | | |
|-------------------------|---------------------|
| -Total Registered Users | -Total Withdrawals |
| -Total Transactions | -Total Transfers |
| -Total Deposits | -Total Bank Balance |

Visual Charts

Transaction trend chart — deposits, withdrawals, transfers over time

Cumulative bank balance chart

Account type distribution chart (Savings / Current / Student)

All KPIs and charts refresh automatically on every admin dashboard load



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

ADDITIONAL FEATURES

Password Recovery

User verifies identity using Username + Account Number + National ID

All three fields must match before password reset is permitted

PDF Bank Statement

Customers download transaction history as a formatted PDF

Filter options: Past 30 days, Past 14 days, Custom date range, Transaction type

Generated in memory — no file saved on server

Account Number Generation

Auto-generated sequentially starting from 301120060001

Unique per user, no manual entry required



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

TESTING SUMMARY

53 Test Cases — All Passed

Module	Test Cases	Result
Authentication	6	✓ Passed
Deposit	5	✓ Passed
Withdrawal	5	✓ Passed
Transfer	6	✓ Passed
Fraud Detection	3	✓ Passed
Fraud Alert Management	4	✓ Passed
Admin Management	6	✓ Passed
Password Recovery	3	✓ Passed
PDF Download	5	✓ Passed
Analytics Dashboard	7	✓ Passed
TOTAL	53	✓ All Passed



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

CHALLENGES & SOLUTIONS

Challenge	Solution
Fraud detection must run before transaction commits	<code>check_fraud()</code> called before every balance update — blocks transaction if fraud detected
Transfer needed atomic updates for both accounts	Single DB connection, single <code>commit()</code> after both sender debit and receiver credit
PDF delivery without saving to disk	Used Python <code>io.BytesIO</code> to generate PDF in memory and stream directly to browser
Admin needed to process blocked transactions post-review	Fraud review handler reconstructs and re-executes original transaction based on stored alert data



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

OBJECTIVES ACHIEVED

Objective	Status
Review and analyze transaction data to generate insights	✓ Achieved
Design and implement a secure banking system	✓ Achieved
Collect and store transaction data for analytical processing	✓ Achieved
Develop an interactive analytics dashboard	✓ Achieved
Develop a rule-based fraud detection system	✓ Achieved
Generate automated reports and downloadable statements	✓ Achieved

All 6 specific objectives successfully achieved



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

LIMITATIONS & FUTURE WORK

Current Limitations

SQLite not suitable for high-volume production environments

Passwords stored without cryptographic hashing

Fraud detection is rule-based only — no ML models yet

Single-branch support only

Future Recommendations

Implement bcrypt password hashing

Integrate ML-based fraud detection using Scikit-learn

Migrate database to PostgreSQL or MySQL

Develop a companion mobile application

Add email notifications for transactions and fraud alerts

Implement Two-Factor Authentication (2FA)



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

CONCLUSION

Smart Banklytics successfully delivers a fully functional intelligent banking system that:

- Processes deposits, withdrawals, and transfers securely
- Detects and responds to fraudulent activity automatically in real time
- Provides administrators with live KPI analytics and trend visualizations
- Generates downloadable PDF bank statements with flexible filtering
- Implements secure role-based access for both customers and administrators

All 6 objectives achieved | 53/53 test cases passed

The project demonstrates the practical integration of software engineering and data science in the financial domain.



BDS 3106 – ONE SEMESTER PROJECT PROJECT PROPOSAL

THANK YOU